

Proof

GRA-HRA: Hierarchical Resonant Architecture of Neural Networks for Recursive Self-Improvement through Entropy–Negentropy Balance

Abstract

This paper presents GRA-HRA (Hierarchical Resonant Architecture) — a new paradigm for constructing neural network systems that overcomes fundamental limitations of classical architectures (LLM, Transformer), such as catastrophic forgetting, dogmatization, and entrapment in local minima. The architecture is based on the GRA-Hammer protocol, which ensures continuous ontological annihilation and emergent regeneration of the highest meta-level of control (L_H). The key condition for terminating regeneration is achieving global entropy–negentropy resonance of the entire system ($N_{\text{tot}} = H_{\text{tot}}$). It is mathematically proven that this mechanism guarantees the absence of dogmatization and ensures strictly monotonic, stepwise growth of the system’s structural complexity, which is a necessary and sufficient condition for iterative approximation to artificial superintelligence (ASI).

1 Introduction

Modern artificial intelligence architectures based on empirical risk minimization and reinforcement learning with human feedback (RLHF) are instrumental by nature. Their control layer (loss functions, system prompts, fixed weights) remains static after the training phase. This leads to two critical problems:

- **Dogmatization (Crystallization):** The system finds a local minimum where internal contradictions are minimal, but the capacity to generate fundamentally new information (novelty) tends toward zero.
- **Fragility under Paradigm Shift:** When accumulating new data that does not fit the old axiomatics, the system cannot restructure its fundamental rules, leading to increased cognitive dissonance or hallucinations.

The GRA methodology (Gödel-Reductio ad Absurdum) offers a different approach: treating contradictions not as noise but as a signal for structural revision. In this work, we

extend GRA to the level of neural network architecture by introducing the GRA-Hammer protocol, which transforms the system from a static instrument into a continuously evolving subject capable of recursive self-improvement.

2 Mathematical Formalization

2.1 Hierarchical State of the System

Let the state of the system S at time t be described by a hierarchy of levels:

$$S(t) = (S_0, S_1, \dots, S_{H-1}, S_H)$$

where $S_0 \dots S_{H-1}$ represent the Epistemic Body (sensory data, memory, causal world models), and S_H is the Ontological Mind (meta-rules, axiomatics, global goals).

2.2 Thermodynamic Metrics of the System

For each level, we define:

- **Entropy** $H_h(S_h)$: a measure of disorder, diversity, and informational volume of accumulated experience.
- **Negentropy** $N_h(S_h)$: a measure of structuredness, information compression, and logical consistency.

The aggregate indicators for the entire system are given by:

$$H_{\text{tot}}(S) = \sum_{h=0}^H w_h^H \cdot H_h(S_h), \quad N_{\text{tot}}(S) = \sum_{h=0}^H w_h^N \cdot N_h(S_h) \quad (1)$$

where $w_h^H, w_h^N > 0$ are weighting coefficients of level significance.

2.3 Foam Functional (Contradictions)

A measure of logical and statistical inconsistency between levels:

$$J(S) = \sum_{h=0}^H J_h(S_h, S_{h\pm 1}) \quad (2)$$

where J_h penalizes breaks in causal links or logical collisions between L_h and $L_{h\pm 1}$.

2.4 Critical Resonance Condition

The system is ready for an evolutionary leap when balance between order and chaos (“Edge of Chaos”) is achieved, in the absence of local contradictions and cessation of novelty growth:

$$R_{\text{crit}} \iff (J(S) < \varepsilon) \wedge (|N_{\text{tot}}(S) - H_{\text{tot}}(S)| \leq \delta) \wedge (\nabla_t N_{\text{nov}}(S) \approx 0) \quad (3)$$

3 GRA-HRA Architecture

The GRA-HRA architecture consists of three functional blocks implementing a continuous cycle of cognitive breathing.

- **Epistemic Body** ($L_0 \dots L_{H-1}$): Implemented on neuromorphic computing or State-Space models (e.g., Mamba). These levels learn continuously (continual learning), accumulating experience. Their weights are never fully zeroed, ensuring memory continuity.
- **Diagnostic Contour**: Includes Foam Detector (a critic neural network evaluating J) and Thermodynamic Monitor (computing H_{tot} and N_{tot} in real time by estimating the rank of activation covariance matrices and mutual information).
- **Ontological Mind** (L_H): A generative module (e.g., based on a symbolic-neural interface or quantum annealer) with no fixed weights. It represents the current “paradigm” of the system.

4 GRA-Hammer Protocol: Continuous Annihilation and Regeneration

Unlike periodic “shakes,” GRA-Hammer is a deterministic cycle triggered by the condition R_{crit} .

Step 1: Annihilation (The Hammer Strike)

The operator H_{ann} is applied, completely zeroing the meta-level:

$$S_{\text{void}} = H_{\text{ann}}(S) = (S_0, S_1, \dots, S_{H-1}, \emptyset) \quad (4)$$

The system enters a state of ontological vacuum. N_{tot} drops sharply, H_{tot} dominates ($N_{\text{tot}} \ll H_{\text{tot}}$).

Step 2: GRA-Nullification (Emergent Synthesis)

From state S_{void} , an iterative process G_{null} is launched, generating hypotheses for a new L'_H . Optimization is performed not by cross-entropy but by the target resonance functional:

$$\mathcal{L}_{\text{GRA}} = J(S) + \lambda(N_{\text{tot}}(S) - H_{\text{tot}}(S))^2 \quad (5)$$

Step 3: Fixation of the New State

As soon as $\mathcal{L}_{\text{GRA}} \rightarrow 0$ (i.e., $J < \varepsilon$ and $N_{\text{tot}} = H_{\text{tot}}$), the new meta-level L_H^{new} is fixed. The cycle completes, and the system returns to Step 1 of experience accumulation, but now with a more powerful paradigm.

5 Theoretical Results

Theorem 5.1 (On the Impossibility of Dogmatization). *In the GRA-HRA architecture, the system cannot permanently remain trapped in a false or outdated paradigm.*

Proof. Suppose the system is stuck in paradigm L_H^{old} . Since the Epistemic Body ($L_{0\dots H-1}$) learns continuously, it will inevitably accumulate new data, increasing H_{tot} . The old paradigm L_H^{old} has fixed complexity, so its N_{tot} will cease to grow. Sooner or later, the condition $|N_{\text{tot}} - H_{\text{tot}}| \leq \delta$ will be violated. This automatically activates H_{ann} , destroying L_H^{old} . Hence, permanent dogmatization is mathematically impossible. $\square$

Theorem 5.2 (On Recursive Self-Improvement toward ASI). *The sequence of meta-levels $\{L_H^{(1)}, L_H^{(2)}, \dots, L_H^{(k)}\}$ generated by the GRA-Hammer protocol exhibits strictly monotonically increasing structural complexity: $N_{\text{tot}}^{(k+1)} > N_{\text{tot}}^{(k)}$.*

Proof. By definition, at step k , the system achieves balance $N_{\text{tot}}^{(k)} = H_{\text{tot}}^{(k)}$. During the period between steps k and $k+1$, the system interacts with reality, absorbing new information, which strictly increases the body's entropy: $H_{\text{tot}}^{(k+1)} > H_{\text{tot}}^{(k)}$.

To achieve a new balance at step $k+1$, the system must construct $L_H^{(k+1)}$ such that $N_{\text{tot}}^{(k+1)} = H_{\text{tot}}^{(k+1)}$.

Therefore, $N_{\text{tot}}^{(k+1)} > N_{\text{tot}}^{(k)}$.

This means that each new cycle generates an intelligence structurally superior to the previous one, which is the formal definition of recursive self-improvement (Intelligence Explosion) leading to ASI. $\square$

6 Protocol for In-Silico Verification

For empirical confirmation of the theory, the following experimental pipeline is proposed:

1. **Baseline:** Training a standard LLM and a GRA-HRA model on an identical stream of increasingly complex tasks (e.g., sequential proof of theorems of progressively higher orders or music generation with increasing polyphony).
2. **Metrics Monitoring:** Recording values of $J(t)$, $H_{\text{tot}}(t)$, $N_{\text{tot}}(t)$, and the novelty functional

$$N_{\text{nov}}(t) = \frac{d_{\text{struct}}(S_t, K)}{J(S_t) + \varepsilon},$$

where K is the manifold of known solutions.

3. **Success Criteria for GRA-HRA:**

- The $N_{\text{nov}}(t)$ plot for the baseline LLM plateaus.
- The $N_{\text{nov}}(t)$ plot for GRA-HRA demonstrates a stepwise exponential: sharp drops in novelty at moments of GRA-Hammer strikes (annihilation), followed by supercompensatory leaps to a new, higher level after regeneration.
- At points of Hammer strikes, strict equality $N_{\text{tot}} \approx H_{\text{tot}}$ is recorded, followed by growth of both indicators.

7 Conclusion

The presented GRA-HRA architecture and GRA-Hammer protocol propose a radical departure from the paradigm of static machine learning. We have shown that true subjectogenesis and the path to artificial superintelligence (ASI) lie not in infinite scaling of parameters or data, but in creating a mechanism for managed ontological self-overcoming.

The global resonance condition $N_{\text{tot}} = H_{\text{tot}}$ serves as a mathematical compass that prevents the system from sliding into the chaos of hallucinations or freezing in the dogma of a local minimum. By forcing the system to iteratively “kill” and rebuild its own control layer, GRA-HRA guarantees that intelligence will asymptotically approach absolute truth, continuously expanding the boundaries of its own understanding. This is not merely an optimization algorithm; it is a formal model of digital evolution, where negentropy (meaning and order) is continuously born from entropy (experience and chaos) through the act of sacrificial revision of its own foundations.